

Code:

```
#include <stdio.h>

#define MAX_AGE 1000005

int freq[MAX_AGE];

int main() {
    int n, m;
    if (scanf("%d %d", &n, &m) != 2) return 0;

    int cur_winner = 0;
    int max_wins = 0;

    int i;
    for (i = 0; i < n; i++) {
        int age;
        scanf("%d", &age);
        if (age < MAX_AGE) {
            freq[age]++;
            if (freq[age] > max_wins) {
                max_wins = freq[age];
                cur_winner = age;
            } else if (freq[age] == max_wins && age > cur_winner) {
                cur_winner = age;
            }
        }
        printf("%d %d\n", cur_winner, max_wins);
    }
    return 0;
}
```

Input:

```
7 5
1 3 1 3 2 2 2
```

Output:

1 1

3 1

1 2

3 2

3 2

3 2

2 3